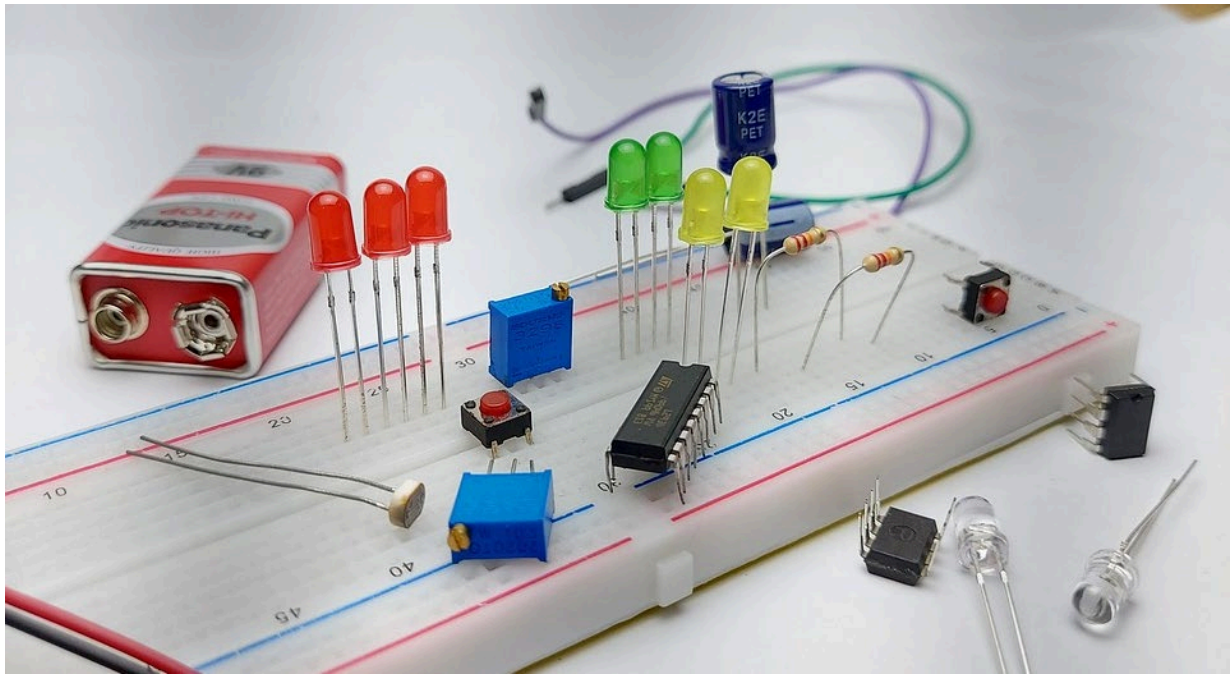


Breadboards

What Is a Breadboard?

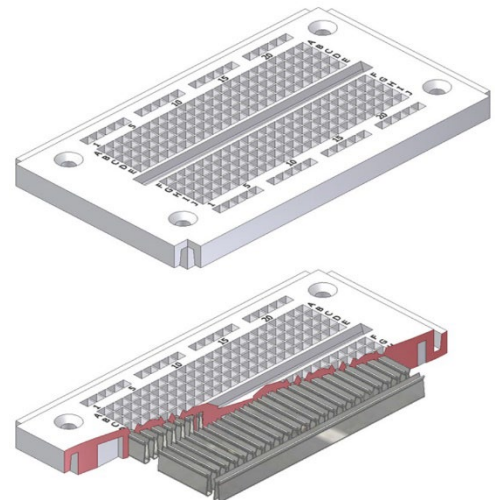
A breadboard, also called a protoboard, is a reusable platform for temporarily building electronic circuits.



As a complement to circuit simulation, breadboarding allows designers to observe how, and if, the actual circuit functions. A breadboard gives designers the ability to quickly change components during development and testing, such as swapping resistors or capacitors of different values. It also enables designers to easily modify a circuit to facilitate measurements of voltage, current, or resistance without soldering.

How a Breadboard Works

Using a breadboard, designers can insert the leads of electric components (like resistors, capacitors, LEDs) or wires into holes that are arranged in a grid pattern on the breadboard surface. Inside the breadboard are a series of metal strips that connect specific rows of holes. Typically, rows and columns are numbered on the breadboard for reference.



Holes are connected in rows and columns as shown in Figure 1. Note that the hole designated as 1a is electrically connected to 1b, 1c, 1d, and 1e. However, 1a–1e are not electrically connected to 1f, 1g, 1h, 1i, or 1j.

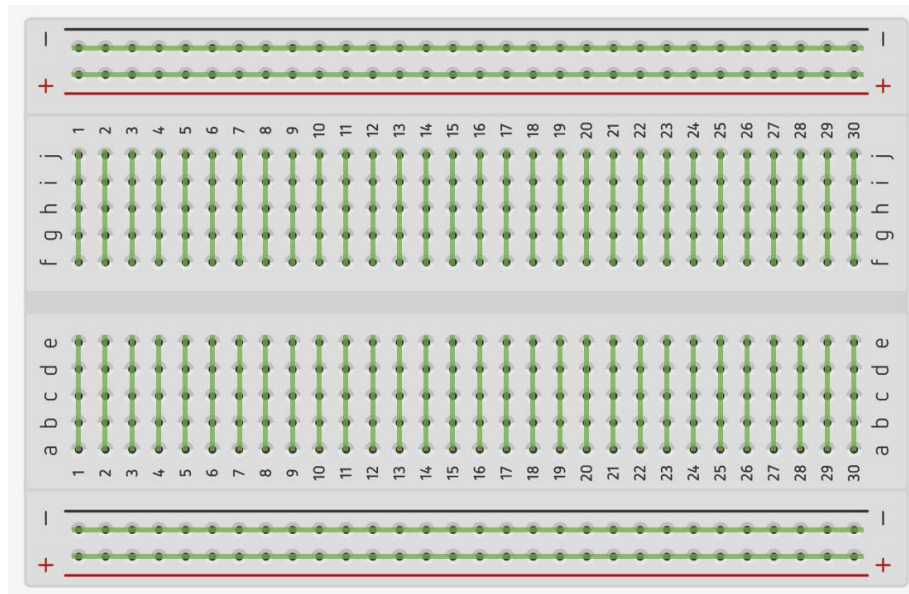


Figure 1. Connected Rows and Columns

Breadboard Guidelines and Tips

When using a breadboard, use as few wires as possible to make your electrical connection. As your circuits get more complicated, having fewer connections can make troubleshooting much easier. Internal breadboard strips should make the majority of connections. When you use wires, keep them as short as possible to avoid jumbled wires, which are difficult to troubleshoot. Compare the circuits in Figures 2 and 3. Electrically, they are identical. But the circuit in Figure 3 would be easier to troubleshoot.

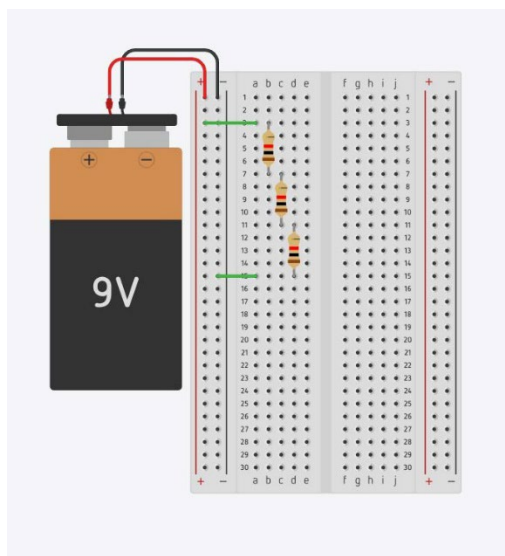


Figure 2. Efficient Breadboarding

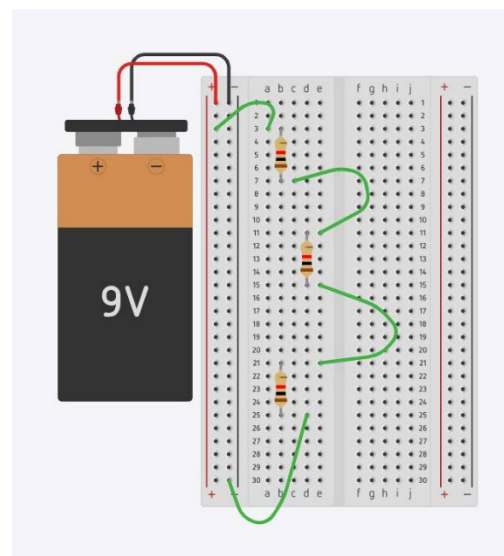


Figure 3. Inefficient Breadboarding